

Designing Critical Edge Case Scenarios for Pedestrian Road Safety

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Abstract—The rapid development of Automated Vehicles (AVs) promises safer and more efficient transportation, but their reliability in real-world conditions, particularly in rare and unexpected edge cases, remains a significant challenge. Pedestrian accidents, which account for nearly 20 percent of road fatalities in Europe, highlight the urgent need for effective automated active safety systems, such as Automated Emergency Braking (AEB), to protect vulnerable road users (VRUs). However, there is currently no universal validation mechanism for testing AVs in such edge scenarios, with existing testing methods relying on predefined sets of cases to address specific requirements. This paper explores methods for testing AVs in critical edge scenarios, focusing on how AVs detect pedestrians in various challenging conditions, including snowy, rainy, and foggy environments, all of which pose risks to automotive safety systems. To evaluate the effectiveness of pedestrian detection under these conditions, the paper conducted a small experiment using the YOLO object detection algorithm. The findings aim to identify vulnerabilities in existing testing processes and contribute to the development of more robust validation methodologies for AVs in critical edge cases.

Index Terms—Automated Emergency Braking (AEB), Vulnerable Road Users (VRUs), Time to Collision (TTC), EURO NCAP, YOLO

I. INTRODUCTION

Autonomous systems in Active Pedestrian Safety are a very crucial technological development in the field of Active Pedestrian Safety Systems. These developments have been ongoing for decades, such as with emergency brakes and automated lane control proposed for car-to-car collisions (Bender et al., 1971) [2], devising control systems for estimating braking power for emergency braking in automated highway systems [3][4]. However, the realization of these systems into automobiles has only recently started, and these gave rise to a standardized test to evaluate and certify auto manufacturers for the effectiveness of these systems. Organizations such as the Euro NCAP are established for this purpose, which lays down test procedures for testing AEB systems for Vulnerable Road Users (VRUs). These test procedures are derived from analysis of real-world accident data. This research was carried out by the Euro NCAP member organizations that played a crucial role in the design of the initial test procedure for the Euro NCAP 2016 and its future revisions [5]. The testing process for AV systems is quite complex, having not only technical trials in testing environments but also on-road testing in a wide range of road and weather conditions including unexpected

and critical scenarios. There are numerous scenarios that AVs may encounter, but it is very difficult to predict and test AVs in all these conditions. This limitation shows the role of Edge Cases in the development of AV.

II. RESEARCH OBJECTIVE

The research aims to enhance the reliability and safety of autonomous vehicle (AV) and active pedestrian safety systems by addressing gaps in existing testing methodologies. It focuses on reviewing Euro NCAP protocols for car-to-pedestrian collisions, comparing them with real-world data to propose improved scenarios that reflect diverse and challenging conditions. Additionally, it evaluates the YOLO object detection algorithm's ability to detect pedestrians in snowy conditions, considering varying clothing types, distances, and lighting. The goal is to develop refined testing standards that better represent real-world scenarios and ensure robust system performance.

III. TESTING SCENARIOS FOR EURO NCAP 2016

There was extensive research done by Colin et al., [5] from Thatcham Research and Apollo Vehicle Safety Ltd based on police-reported crash data covering Great Britain [1] that led to the following findings:

- Single largest group of pedestrians involved crossing from the nearside during daylight with average vehicle speed of 43 kmph of which only half braked with speed reduction up to 7 kmph.
- Next largest group formed of children crossing during daylight from behind an obstacle.
- Third largest cluster comprised of pedestrians crossing from both far-side and nearside in dark / wet weather conditions. Vehicle speeds in such cases were always over 50 kmph and included the highest number of fatalities.

Based on these findings a summarized version is presented in Figure 1.

A. Dummy Selection

The development of pedestrian test targets, detailed in a previous paper (Lemmen et al., 2013 [15]), focused on factors such as dimensions and body stance, infrared reflectivity, radar cross section (RCS), and background contrast. A low-profile platform was chosen for propulsion to reduce visual and radar signatures. The 4A dummy became the preferred option

Combining accident data from international sources		UK	UK	Germany	USA
		STATS 19 n=10,574 cluster analysis frontal collisions	OTS n=175 cluster analysis frontal collisions	UDV n=234 (n=18,571) 3rd party vehicle claims 2002- 2006 frontal collisions	IIHS 1997-2006 FARS & GES all car pedestrians
Pedestrian walks from nearside		51%	59%	32%	27%
Pedestrian walks out from behind obstruction		14%		7%	
Pedestrian runs out from the far side		9%	37%	28%	
Pedestrian walks along in the dark		3%	5%	8%	9%
Pedestrian walks out into the path of turning car		6%	Overall: going ahead 87%, Turning 13%	18%	-

Fig. 1. International comparison of the frequency of different pedestrian crash types Note: By Colin et al., [5]



Fig. 2. 4A Initial Test Target without contrast Note: By Colin et al., [5]

after extensive development and testing, showing robustness at 60 km/h impacts with minimal damage. Visual recognition was good, but radar performance varied, particularly with the Mercedes E-class. Workshops in 2014 refined the radar signature and clothing color, selecting blue trousers, a black top, and black hair for optimal contrast.



Fig. 3. 4A revised test target Note: By Colin et al., [5]

B. Test Scenarios

The following test conditions were evaluated and tested based on the accident data [3]:

1) **Target walks from nearside, no obstruction:** The testing scenario involved a target walking at 5 km/h and vehicles operating at speeds of up to 60 km/h, measured using Time to Collision (TTC). While the theoretically greater reaction time

offered advantages, limitations arose due to the restricted field of view of sensors, particularly at lower vehicle speeds. This affected the overall performance of the system in detecting and responding effectively to the pedestrian.

2) **Target walks from far side, no obstruction:** The scenario describes an autonomous vehicle system where a pedestrian dummy is walking at 5 km/h. Initially, the system has low confidence in identifying the dummy as a pedestrian until it enters the vehicle's direct path. Despite this uncertainty, the system's decision-making time for initiating braking remains consistent with previous cases.

3) **Target runs from far side, no obstruction:** The scenario describes a system where the target speed is 8 km/h. The impact point is considered at 50% of the vehicle's width from the nearside. Advanced radar and camera fusion systems can detect obstacles up to 55 km/h and slow the vehicle to 5 km/h. However, systems without path prediction algorithms experience longer decision-making times before Time to Collision (TTC), leading to slower responses.

4) **Elderly adult walks slowly from farside:** At a target speed of 3 km/h, testing at slow speeds can be challenging due to difficulties in detecting or quantifying the vehicle's response and performance.

5) **Child runs out from behind parked vehicles at the nearside:** At a target speed of 8 km/h, an obstruction scenario involves a smaller car in front of the pedestrian, with a larger car in front of that. Due to the lower Time to Collision (TTC), impact points were considered at 25% of the vehicle's width from the nearside. The lower reaction time for sensors to detect the target and make decisions posed challenges for effective braking.

C. Final Scenario Selection

Considering the limitations for different scenarios, the summarized final test scenario (Fig-4) was suggested to Euro NCAP 2016 which only listed cases that were technically feasible:

PEDESTRIAN				
	Far side		Near side	
	Adult		Adult	Child
	8km/h		5km/h	5km/h
	50% (central)		25% & 75%	50% (central)
VUT	20-60km/h		20-60km/h	20-60km/h

Fig. 4. Test scenarios to be incorporated Euro NCAP 2016 Note: By Colin et al., [5]

The test scenarios were tested with two vehicles: Mini One (equipped with single mono-camera AEB) and Lexus LS460 (equipped with fusion sensor system of radar, stereo camera, infrared sensors). The Mini could perform only for nearside scenarios whereas the Lexus could perform in all scenarios except when the target is hidden behind an obstacle and

emerging at the last 2 seconds giving it less time to react. Euro NCAP incorporated these scenarios and published a new protocol called as AEB/LSS VRU systems. The final testing scenarios are shown in Fig-5.

D. Real Accident Data

The Facts and Figures [8] is a report summarizing the road-safety incidents on European roads and is supervised and published by the European Commission. The current version of 2024 references from 2022 statistics of pedestrian accident data. The following observations were summarized based on their report:

1) Based on Gender and Age:

- The share of men among pedestrians is lower as compared to women which constitute 38 percent of pedestrian fatalities.
- Half of all pedestrian fatalities involved senior citizens (65+ year olds) from 2022 as compared to other road users (Fig-6 and 7)

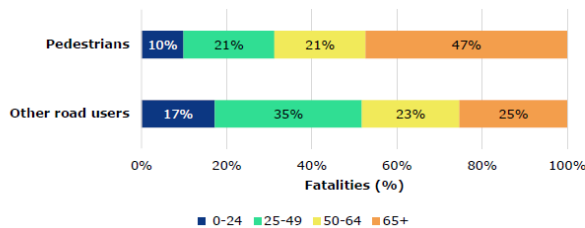
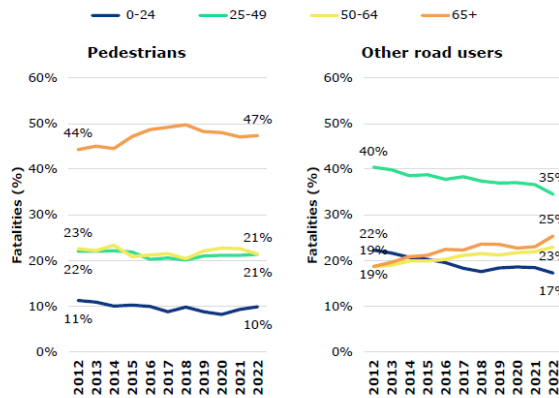


Fig. 5. Pedestrian fatalities by age Note: From Facts and Figures Pedestrians [8]



In some European countries, a relatively high share of seniors among pedestrian fatalities (between 56 and 62%) is observed, such as in Greece, Slovenia, Italy, Spain, Germany and Finland.

Fig. 6. Pedestrian fatalities trend 2012-2022 Note: From Facts and Figures Pedestrians [8]

2) Based on Location and Track Conditions:

- Pedestrian fatalities occurred most on urban roads followed with rural roads and motorways. However, coun-

tries such as Latvia, Sweden and Norway, fatalities occurred more on rural roads.

- Almost 1 out of 5 pedestrians got fatally injured at road junctions
- Surface conditions were dry for 78 percent of pedestrian fatalities, while it was wet for 20 percent and snowy for 1 percent (Fig-8)

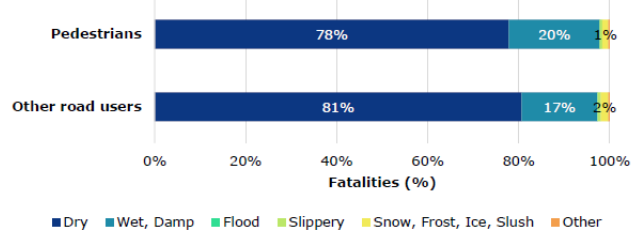


Fig. 7. Pedestrian fatalities based on location and road conditions Note: From Facts and Figures Pedestrians [8]

3) Based on Light Conditions and Time of Day and Year:

- Pedestrian fatalities occur frequently during darkness as compared to other road users (Fig-9)
- Pedestrian fatalities occur more often at daytime during weekdays but more often at nighttime during weekends
- Pedestrian fatalities peak during the winter months from December to January (Fig-10)

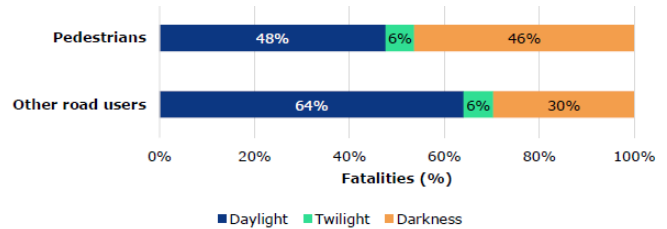


Fig. 8. Pedestrian fatality trend for time of day Note: From Facts and Figures Pedestrians [8]

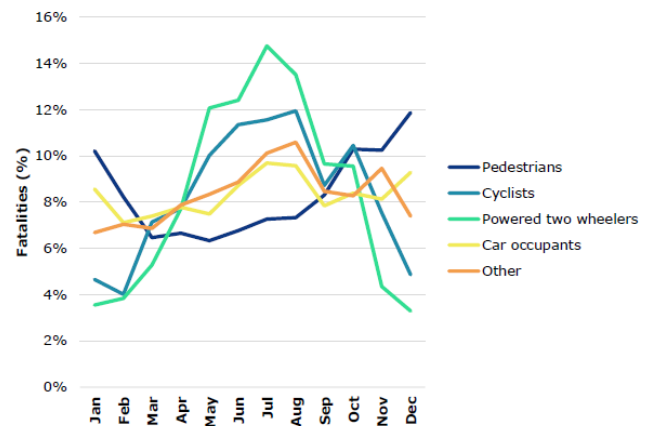


Fig. 9. Pedestrian fatality trend for time of year Note: From Facts and Figures Pedestrians [8]

IV. OBSERVATIONS

The current protocol's [7] following testing criteria clearly shows gaps with the real accident findings as stated above:

- Current European Pedestrian Target (EPT) uses only child (EPTc) and adult (EPTa) as per ISO 19206-2:2018 [9]
- Test track conditions currently focus on the following scenarios [7]:
 - 1) Dry conditions with no visible moisture on surface
 - 2) Surface to be paved with no irregularities (cracks, manhole covers, large dips, etc.)
 - 3) Ambient temperature between 5°C and 40°C
 - 4) No precipitation should be falling
 - 5) Night time conditions testing to be carried under street lighting of 15 lux
- The limitations of the current scenarios are as follows:
 - 1) Senior pedestrians targets are not represented in the European Pedestrian Target even though they fall in significant percentage of road fatalities
 - 2) Rain and Snow are not considered for track conditions
 - 3) Current track conditions are representative of an urban scenario. Rural roads with track unevenness are not taken into account. This does not entail some European countries where accidents are more prevalent in rural areas.
 - 4) Night-time conditions with twilight and darkness are not evidenced in the protocol, even though half of pedestrian accidents occurred during darkness.

V. LITERATURE SURVEY

Based on the limitations presented above, this section would delve into parallel research done in this field and highlight whether current technical advancements in autonomous driving are capable to perform under harsh conditions which is representative of the limitations presented earlier.

A. Rainy Conditions

- The experimental setup of Hasirlioglu et al.,2016 [10] worked with RADAR, LIDAR and Camera as automotive surround sensors to detect object under rain conditions of 94.5 mm/h. The goal of the research was to study the effect of rain on these sensors and how it affects their performance. An indoor rain frame setup was devised with four layers of rain that could be controlled effectively and imitated natural rain. For RADAR and LIDAR it was found that the electromagnetic and light waves respectively decreased in intensity with increase of rain layers. The standard long range 77GHz radar finder gave a characteristic RCS Value Range for each object type and under rain these values got lowered leading to mixing of value ranges and detecting incorrect objects. In case of LIDAR, the light wave due to repeated absorptions and reflections got scattered which resulted in the received wave having less intensity than released. This often resulted in the sensor detecting multiple ghost

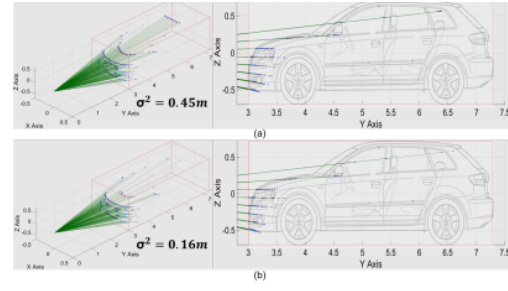


Fig. 10. Comparison of points dispersion and extrapolation to 3D Object a) with rain and b) without rain Note: From Vriesman et al.,2020 [11]

objects instead of the single target. Similar experimental study conducted by Vriesman et al.,2020 [11] concluded with both RADAR and LIDAR showing drop in intensity with rain as compared to without rain. But with LIDAR they were able to concentrate the position of the object by two methods:

- 1) First method was by estimating the position of the target through known values, a bounding box was created inside which the distance of the points detected by LIDAR were calculated and normalized. The quantity of points dropped with increase in distance.
- 2) Second method used euclidean distance of the points inside the bounding volume box and variance is applied for each frame. Finally, average of all the variances are taken and extrapolated to a 3D figure as shown in Fig-10.

- These experiments were able to show the performance of the surround sensors in rainy condition and demonstrate the extent to which the performance can deteriorate in rainy conditions. This gives the research community information on what to expect and finding better algorithms that can be incorporated to improve the performance of such sensors in extreme conditions.

B. Slow Moving Pedestrians

- Significant research have been conducted in the field of detecting slow-moving pedestrians as targets. One such report by Andres et al., 2012 [12] talks about signal post-processing of radar signals. The researchers used a technique of "Chirp Sequence Processing" that helped in amplifying the bandwidth of the radar signal using linear prediction. A series of tests were carried out where a pedestrian walking at 4.5 km/h from the side of the car was detected by a radar sensor. First test was carried with a bandwidth of 0.5 GHz and the separation cluster between the car and pedestrian was low. The second series of tests were carried out with a bandwidth of 1.35 GHz and in this case better separation could be achieved. The final test was carried out with linear prediction with 0.5 GHz bandwidth in which the clusters were extrapolated and a velocity resolution of 0.45 km/h was

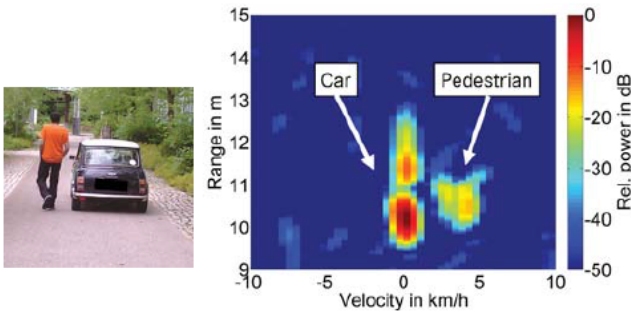


Fig. 11. Linear prediction of radar signal with separation between car and pedestrian Note: From Andres et al., 2012 [12]

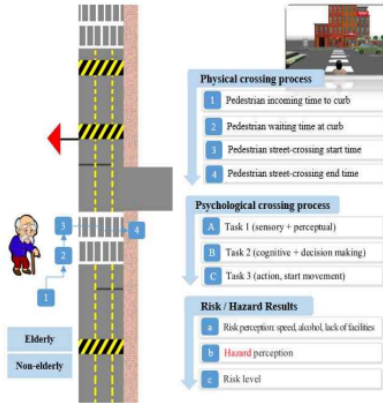


Fig. 12. Test Methodology from Jang et al., 2016 [13]

achieved with a better separation between the car and the pedestrian (Fig.11).

- Another set of studies by Jang et al., 2016 [13] takes a different approach for early detection of elderly pedestrians. This study uses a combination of Remote server, mobile devices and a route planning server. First a series of data is collected such as the elderly pedestrian's walking behavior and data logging. This data is fed into an image data processing server that calculates the pedestrian's heading direction in real time. And finally, the remote server analyzes the large sample of collected data to develop an analysis report on the walking pattern of the elderly person (Fig.12). This approach calls for an intelligent communication system between car-to-pedestrian where both being connected to Information sharing services (ITS) can exchange real-time data, which can help in collision avoidance based on both the car's and pedestrian's velocity and trajectory. Although this method has not yet been field tested, verification of this test procedure will occur in the future.

C. Foggy Conditions

- Guan et al., 2020[16] study investigates the use of millimeter wave radar for high-resolution imaging in foggy conditions, where traditional vision-based systems often fail. The technology leverages the high-frequency proper-

ties of millimeter waves to penetrate dense fog, ensuring reliable detection and imaging. Key findings highlight its potential for enhancing safety and navigation in adverse weather conditions, particularly in automotive and transportation applications. The study discusses radar's advantages, including weather resilience, and addresses challenges like resolution limitations and interference. Experimental results validate the effectiveness of millimeter wave radar for robust imaging under low-visibility scenarios, underscoring its critical role in advancing autonomous and safety-critical systems.

- Li et al., 2019[17] deals with the enhancement of deep learning-based models of pedestrian detection in haze weather by solving the problems of visibility degradation. Three models, which employ the You Only Look Once (YOLO) algorithm at its core, were created with the implementation of depth wise separable convolution, linear bottleneck, and a novel weighted mixture layer. These changes aimed to improve the models' efficiency and reduce their computational cost. Such models exceed conventional techniques in both accuracy and speed. The training data was improved by employing dataset augmentation techniques such as horizontal flips, contrast changes, cropping, brightness changes, affine transforms, addition of Gaussian noise, and blurs, among others. The models were verified experimentally with average precision, recall and frame rate metrics to showcase strong pedestrian detection capabilities in hazy environments.
- M. Bijelic et al. [18] discuss object detection in adverse weather conditions such as dense fog, snow, and rain. It comes with a unique multimodal dataset of over 10,000 km collected in northern Europe using various sensors: lidar, radar, RGB cameras, gated NIR, and FIR cameras with over 100,000 labeled instances. These conditions skew sensor streams in an asymmetric way that degrades detection accuracy. The authors introduce an adaptive deep fusion network based on measurement-entropy capable of fusing heterogeneous sensor features and robustly detecting extreme weather in the absence of large labeled datasets. Adaptive feature fusion contributes to its performance gain, which is realized in the case of dense fog. Such work lays the groundwork for resilient sensor fusion in autonomous vehicles.

VI. PROPOSED TESTING SCENARIOS

Based on the limitations presented in the current protocol[7], a new testing scenario was thought that incorporates the current technical possibilities to test under such conditions. A brief description of the proposed test setup could be as follows.

- Standard ISO 19206-2 [9] dummy specifications as per Annex A.
- Visually dummy should have two hair types, black and white. Brown full-sleeve T-shirt, black trousers and brown shoes are suggested.

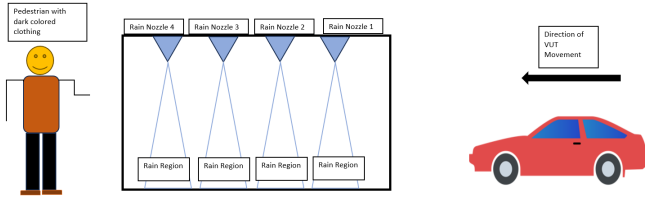


Fig. 13. 2D Schematic of test setup with Rain Simulator

- Mobile rain simulator setup with adjustable height (between 3m and 5m) and area of 5m X 2m.
- Simulator should have four equally spaced rows of nozzles. Number of nozzles in a row can be decided by the user. Each nozzle should sprinkle steady stream of water such that the final effect imitates real rain condition. A 2D visual schematic is shown in Fig-13.
- A final Car-to-Pedestrian testing scenario is devised using the format from [7]. The scenario consists of four scenarios:
 - **Car-to-Pedestrian Nearside Adult (CPNA)** : As defined by [7] for 25/75% of the car's width.
 - **Car-to-Pedestrian Farside Adult (CPFA)** : As defined by [7]
 - Daylight scenario with wet track conditions
 - Night conditions with both dry and wet track conditions (Fig-14)

VII. REAL WORLD TESTING USING YOLO

This section details the methodology for testing the robustness of YOLO (You Only Look Once) object detection algorithm in automotive safety. The testing is designed to simulate critical edge case scenarios by adding different clothing types, environmental conditions (day/night), and snowy weather conditions and varying distances.

A. Test Setup

- **YOLO Algorithm Configuration:** YOLOv11 is selected for its high accuracy and real-time processing capabilities. YOLOv11 is particularly suited for scenarios where rapid and precise detection of objects is critical, ensuring efficient handling of varying environmental and object-based challenges.
- **Target Clothing:** The test focuses on detecting individuals wearing low-contrast jackets and trousers. These outfits are specifically chosen to mimic real-world scenarios on a snowy day, where the clothing blends into the environment. The selected attire includes a combination of grey, black, and white jackets, paired with trousers of similar tones. This setup challenges YOLOv11's ability to detect objects with minimal contrast against the snowy background.
- **Lighting Conditions:** The testing environment accounts for two distinct lighting scenarios to evaluate YOLOv11's performance under diverse illumination:

- 1) **Daytime:** Testing takes place in natural daylight, which includes conditions such as overcast skies and shadows cast by nearby objects. This helps simulate real-world variations in daytime lighting.
- 2) **Nighttime:** Low-light conditions are simulated with varying streetlight arrangements. These may include areas with adequate lighting, dimly lit sections, or unevenly distributed light sources.

• Test Environment:

- 1) **Location:** The experiments are conducted in an open and controlled area chosen for its accessibility and ability to simulate various real-world conditions in Deggendorf.
- 2) **Camera:** The camera used for testing is an iPhone 16, mounted on a sturdy tripod for stability during data collection.
- 3) **Weather Conditions:** Testing takes place under snowy weather conditions, which introduce additional challenges for object detection. The conditions include both Falling snow, which may obscure visibility and create visual noise and Piled-up snow, which reduces contrast between the target objects (clothing) and the environment.
- 4) **Varying Distance:** The tests are conducted across a range of distances, from 5 meters to 25 meters

Figures 14 to 19 illustrate various test scenarios conducted as part of the study. These scenarios encompass different environmental conditions, clothing types, and distances to evaluate the robustness of the YOLO object detection algorithm.

TABLE I
YOLO OBJECT DETECTION PERFORMANCE

Distance (m)	Day Confidence Score			Night Confidence Score		
	White	Black	Gray	White	Black	Gray
5	0.89	0.91	0.89	0.92	0.90	0.91
10	0.86	0.84	0.82	0.80	0.84	0.80
15	0.70	0.72	0.75	0.69	0.80	0.66
20	0.68	0.69	0.61	0.48	0.67	0.67
25	0.71	0.63	0.55	0.36	0.46	0.42

VIII. TEST RESULTS AND DISCUSSIONS

Table 1 provides a comparison of YOLO object detection algorithm's performance in two distinct lighting conditions: daytime and nighttime, with varying object colors (white, black, and gray) and distances (from 5 meters to 25 meters). Below, we will discuss the algorithm's performance in both lighting conditions in detail, breaking down how these factors influence detection accuracy. A. Daytime Performance: Fig 8 showcases the relationship between distance and confidence scores of the object detection algorithm in daytime conditions. The confidence scores, which range from 0 (no confidence) to

TABLE 2. PROPOSED NEW TEST SCENARIO

Scenario	CPNA	CPFA	CPNA	CPFA	CPNA	CPNA	CPNA	CPNA
VUT Speed (kph)	10-50	10-50	10-50	10-50	10-50	10-50	10-50	10-50
Target Speed (kph)	5	8	3~4.5	3~4.5	5	3~4.5	5	3~4.5
Track Conditions	Wet	Wet	Wet	Wet	Dry	Dry	Wet	Wet
Rain Intensity	Nozzle Series*	Nozzle Series*	Nozzle Series*	Nozzle Series*	Nozzle OFF	Nozzle OFF	Nozzle Series*	Nozzle Series*
Target Direction	From Nearside	From Farside	From Nearside	From Farside	From Nearside	From Nearside	From Nearside	From Nearside
Impact Location	25/75% of vehicle's width	50% of vehicle's width	25/75% of vehicle's width	50% of vehicle's width	25/75% of vehicle's width	25/75% of vehicle's width	25/75% of vehicle's width	25/75% of vehicle's width
Lighting Conditions	Daylight	Daylight	Daylight	Daylight	Nighttime (No Streetlights)	Nighttime (No Streetlights)	Nighttime (No Streetlights)	Nighttime (No Streetlights)
Vehicle Lights	OFF	OFF	OFF	OFF	ON (high beam)	ON (high beam)	ON (high beam)	ON (high beam)

Note: * Nozzles to be opened in succession (First test with only Nozzle 1, second test with Nozzle 1 & 2 and so on)



Fig 14. Detection of a person in a black jacket at 5m during daytime



Fig 16. Detection of a person in a gray jacket at 5m during daytime



Fig 15. Detection of a person in a white jacket at 5m during daytime

1 (high confidence), reflect the certainty of the algorithm in successfully detecting the object. The data reveals a general trend of declining confidence scores as the distance increases, highlighting the challenges faced by the algorithm in detecting objects farther from the camera due to reduced object size, lower pixel resolution, and diminished contrast. Among the



Fig 17. Detection of a person in a black jacket at 5m during nighttime



Fig 18. Detection of a person in a white jacket at 5m during nighttime



Fig 19. Detection of a person in a gray jacket at 5m during nighttime

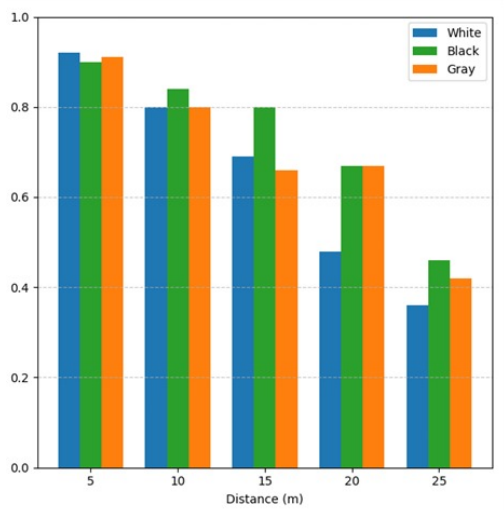


Fig 20. Confidence Score v/s Distance for Night

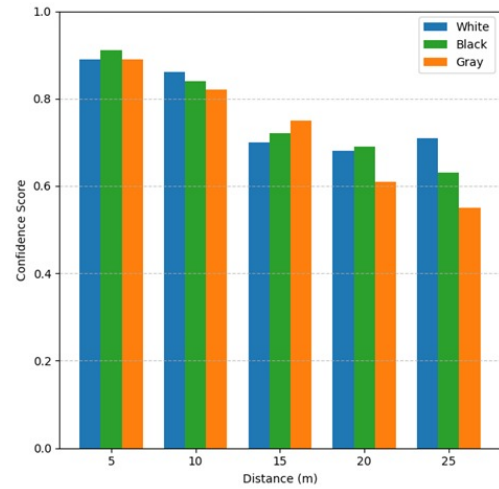


Fig 21. Confidence Score v/s Distance for Day

three clothing colors, white clothing consistently outperforms black and gray, with confidence scores remaining relatively high and stable across all distances. Starting with a score of 0.89 at 5 meters, the detection confidence for white decreases gradually, reaching 0.68 at 20 meters and recovering slightly to 0.71 at 25 meters. This can be attributed to white clothing's strong light reflectivity and higher contrast against most backgrounds, which enhances its visibility and detectability by the algorithm. Black clothing, on the other hand, shows strong initial performance with the highest score of 0.91 at 5 meters but experiences a steady decline as the distance increases. By 20 meters, the confidence drops to 0.69, and 0.63 at 25 meters. Black clothing absorbs more light and provides less contrast, it becomes progressively harder for the algorithm to distinguish at greater distances. Gray clothing emerges as the most challenging color for the algorithm to detect, with a confidence score that starts at 0.89 at 5 meters but deteriorates significantly as the distance increases. By 15 meters, the score drops to 0.75, and by 25 meters, it reaches the lowest value of 0.55 among all tested scenarios. Gray clothing's moderate reflectivity and neutral contrast often blend with various backgrounds, especially in suboptimal lighting conditions, resulting in a dramatic decline in detection performance.

B. Nighttime Performance: The performance of the object detection algorithm at night varies significantly across different clothing colors—white, black, and gray—highlighting distinct trends and challenges at various distances, as shown in Figure 9. White clothing while initially performing strongly due to its reflective properties, shows a more pronounced decline in detectability as distance increases. At 5 meters, white clothing achieves the highest score of 0.92, reflecting excellent detectability at close range. However, this advantage diminishes quickly as distance grows. By 10 meters, the score drops to 0.80, and by 15 meters, it falls further to 0.69, indicating a significant loss in detectability compared to black clothing. At

20 meters, white clothing experiences a sharp decline to 0.48, and at 25 meters, it matches gray clothing at a low score of 0.36. The steep drop in confidence suggests that while reflective properties provide an advantage at close ranges, they become less effective at longer distances in low-light conditions. Black clothing is the most consistently detectable color across various distances, contrary to expectations. At 5 meters, black clothing achieves a high confidence score of 0.90, slightly below white (0.92). Remarkably, as distance increases to 10 meters, black clothing maintains a confidence score of 0.84, outperforming white clothing, which drops to 0.80. This trend continues at 15 meters, where black clothing achieves a score of 0.80, significantly higher than white (0.69) and gray (0.66). Even at 20 meters, black clothing maintains a respectable confidence score of 0.67, again surpassing white (0.48) and matching gray. At the furthest distance of 25 meters, black clothing retains the highest detectability among all colors with a confidence score of 0.46, while white and gray plummet to 0.36 and 0.42 respectively. These results indicate that, surprisingly, black clothing proves to be the most consistently detectable color under nighttime conditions, likely to benefit from specific contrasts with certain dark or dimly lit backgrounds. Gray clothing consistently demonstrates the weakest performance among the three colors under nighttime conditions. At 5 meters, gray clothing achieves a high confidence score of 0.91, similar to the other colors, but its detectability rapidly declines as distance increases. By 10 meters, the confidence score falls to 0.80, matching white clothing but lagging behind black. At 15 meters, gray clothing exhibits the lowest score of 0.66, reflecting the algorithm's difficulty in distinguishing this color from the background. At 20 meters, gray clothing stabilizes at 0.67, matching black clothing, but by 25 meters, it drops significantly to 0.42.

IX. CONCLUSION

Thus, there are conspicuous lacks in the available testing of automotive sensors under extreme conditions; hence, there is a test protocol to try and exploit real accident data from the real world. The following examines the capability of the YOLO object detection algorithm in detecting pedestrians in snowy conditions. It ranges in finding a decrease in confidence with distance, white works the best during daytime, black in the dark, and gray being the toughest condition. This indeed means that far more adaptive testing of corner cases, like snowy conditions, is called for to make sure the autonomous vehicles can operate safely under any condition.

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